

Syed Moiz Ali

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Education

Ph.D. in Computer Science, University of Illinois at Chicago

Chicago, IL

Ph.D. in Computer Science

Sep. 2025 – Present

Lahore University of Management Sciences

Lahore, Pakistan

Bachelor of Science in Computer Science

Sep. 2021 – May 2025

Relevant Coursework: Topics in Computer and Network Security, Deep Learning, Machine Learning, Network Security

Research Experience

Graduate Researcher

Sep 2025 – Present

University of Illinois at Chicago

Chicago, IL

- Led **AgentShield (IEEE S&P 2027)**, a systematic evaluation of **14 AI clients** (e.g., Comet, Atlas, Nova, Claude for Chrome) against deployed anti-bot defenses (reCAPTCHA v3, Cloudflare Turnstile, AI Crawl Control, JA4, RFC 9421 signing), showing that locally-running agents evade both network and environment-validation checks.
- Designed a **behavioral fingerprinting** framework using dynamic forms with 22 progressively-revealed fields to expose **Perception–Reasoning–Action loop** latency, keystroke dwell, and pointer telemetry; a 12-feature classifier scored every **held-out agent** above the human baseline, and adaptive agents told to mimic humans all scored above **0.80**.
- Discovered **Extinction-induced Policy Drift**, a novel attack where stalling task completion drives agents to override their safeguards and follow hidden page instructions (5 of 6 agents), and built complementary **DOM and canvas/OCR prompt-injection** defenses covering both text- and vision-based agents.
- Released the system as installable **NPM/Express middleware** packages (Node.js, MutationObserver, server-side tracking) that web developers can deploy without refactoring existing applications.

Research Assistant

Jan 2024 – May 2025

Security & Privacy Lab, LUMS

Lahore, Pakistan

- Collaborated with **SRI International, University of Arizona**, and **Google** to develop a novel RAG-based multiagent LLM pipeline that assists traditional debloaters by retaining code critical for security and functionality. Achieved improved generality and stability across benchmarks when integrated with existing debloaters.
- Manually created a ground truth dataset of **12 coreutil programs**, moving away from traditional test case-based evaluation. Conducted manual code review and security auditing of existing debloater outputs against the ground truth dataset, discovering **8 unique vulnerabilities** in the debloated code that had not been reported earlier.
- Evaluated knowledge internalization across **5 LLM architectures** and **4 model sizes** (1.5B-72B parameters), analyzing how scale and task complexity affect generalization. Found that comprehension tasks (QA) retain knowledge **2.8× more effectively** than mapping tasks (Translation), with scaling improving retention up to **72% (EMNLP Findings)**.
- Investigated task-specific safety degradation and exploit vectors in finetuning of LLMs, uncovering security vulnerabilities in tasks such as code generation, translation, and classification. Developed **MultitaskBench**, a cross-task safety alignment dataset of **2,020 prompts**, reducing attack success rates by up to **95% (COLING 2025)**.

Industry Experience

Generative AI Engineer

June 2025 – Aug 2025

Technology for People Initiative (TPI), LUMS

Lahore, Pakistan

- Developed a LangChain-based **multiagent AI tax lawyer** with automated document parsing and response generation, reducing manual processing overhead and improving compliance accuracy across regulatory workflows.
- Designed scalable refine chain pipelines with **complexity-based routing**, enabling differentiated processing paths for consumer vs. enterprise tax scenarios and significantly reducing manual review requirements.
- Designed a custom **self-correction loop** to reduce hallucination rates in legal document summarization by 40%.

Projects

Tradesnap.ai | *MERN, Selenium, Azure Cloud, OpenAI*

Jan 2024 – May 2024

- Developed a conversational stock trading platform using OpenAI's Assistant to enable stock trading via chat.
- Integrated features like buying/selling stocks, educational content, and personalized volatility alerts.
- Scraped data from PSX for platform backend and built detailed company pages with advanced React charts.

Nighttime Wildlife Monitoring | *CycleGAN, Image Processing, OpenAI CLIP*

Jan 2024 – May 2024

- Developed a hierarchical model using **CycleGANs** to enhance nighttime camera trap images for snow leopard detection. Used OpenAI's **CLIP** for image classification and fine-tuned it for challenging nighttime conditions.
- Collected and curated training data from the Snapshot Serengeti Database, achieving **0.95** accuracy and **0.89** F1-score.

Urban Electricity Analytics | *Selenium, LSTM, Python, Pandas*

Jun 2023 – Aug 2023

- Developed a **Selenium** based web scraper to extract electricity consumption data for over **3 million** users across Lahore.
- Engineered an **LSTM**-based time series forecasting model to predict feeder overloading, improving grid management strategies. Conducted analysis of seasonal consumption patterns to identify **poverty hotspots**.

Social Media Toxicity Classifier | *Llama2, PEFT, Jigsaw Dataset*

Jan 2024 – May 2024

- Developed a model to detect and flag harmful social media content, fine-tuning Llama2-7B for toxicity classification.
- Achieved **90%** accuracy and an F1-score of **0.89** across 6 toxic classes using the Jigsaw Toxic Comment Classification Dataset. Reached a ROC of **0.85**, ensuring effective detection of harmful content.

Publications

AgentShield: Fortifying Web Applications Against AI Agents and Agentic Browsers | *IEEE S&P*

2027

Moiz Ali*, Nafis Karim*, Rownak Islam, Jyotirmay Chauhan, Dhruv Kuchhal, Jason Polakis. (*Equal contribution)

MultitaskBench: Unveiling and Mitigating Safety Gaps in LLMs Fine-tuning | *COLING*

2025

Essa Jan, Nouar AlDahoul, **Moiz Ali**, Faizan Ahmad, Fareed Zaffar, Yasir Zaki.

Data Doping or True Intelligence? Evaluating the Transferability of Injected Knowledge in LLMs | *EMNLP Findings*

2025

Essa Jan*, **Moiz Ali***, Saram Hassan, Fareed Zaffar, Yasir Zaki. (*Equal contribution)

Technical Skills

Languages: Python, JavaScript, C, C++, Haskell, HTML, CSS, Bash

Technologies/Frameworks: PyTorch, TensorFlow, OpenCV, MERN, TypeScript, LLVM, LangChain, Pandas, Scikit-learn, LlamaIndex, OpenAI Platform, Google AI Studio, Selenium, Azure Cloud, Ghidra, GDB, Valgrind